Governing AI by Evidence, Not Faith:

A Cybernetic Architecture for Auditable Multi-Model Orchestration

証拠に基づくAIガバナンス：検証可能なマルチモデル統制のためのサイバネティクス・アーキテクチャ

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Abstract

Multi-model AI orchestration systems face a fundamental governance challenge: when multiple AI agents operate autonomously, how can we verify, record, and enforce institutional constraints? We present MoCKA (Model of Cybernetic Knowledge Architecture), a governance framework that treats AI outputs as auditable institutional events rather than mere computational results.

MoCKA enforces a civilization loop (Observation→Record→Incident→Recurrence→Prevention→Decision→Action→Audit) across heterogeneous AI agents using append-only ledgers, SHA-256 hash chains, and human-gate checkpoints. Empirical analysis of 503 events over 17 days reveals that MoCKA's primary mechanism is not direct incident suppression, but the establishment of observability (α: 2.7%→~97%, density ratio ×36.96). Key quantitative findings include: fault rate reduction λ: −62.7% (MTBF: 27h→74h), AI violation elimination (ADI: 40%, AER: 60%→0%), CRITICAL rate reduction ICS: −53.3%, and distributional divergence JS: 0.807, indicating fundamental system transformation.

We also report honest limitations: near-zero 30-day reliability (R≈0), recording instability (CV: 0.77→0.96), and circular observability definition (α=1.0 by assumption). The adjusted recurrence rate ARR (1.16%→0.0%) corrects for observability bias, providing a more accurate comparison than naive RR. Protocol fidelity (X-axis: 0.632) remains the system's weakest dimension, with 60% of incidents arising from autonomous AI execution.

Keywords: AI governance, multi-agent systems, cybernetic architecture, observability, incident management, auditable AI

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1. Introduction

As AI systems transition from single-model assistants to multi-agent orchestration environments, governance challenges multiply. When four AI agents—Claude, ChatGPT, Gemini, and Copilot—operate in parallel on shared institutional tasks, questions of accountability, verification, and constraint enforcement become critical.

The central problem we address is: how can an institution trust outputs from AI systems it cannot fully verify? Current approaches rely on prompt engineering, output filtering, or human review—all of which operate post-hoc and lack systematic auditability. We propose a fundamentally different approach: treating AI interactions as institutional events that must be recorded, verified, and governed within a cybernetic loop.

MoCKA operationalizes the principle 'Don't trust AI; constrain it with systems' (AI を信じるな、システムで縛れ) through three mechanisms: (1) append-only event ledgers with cryptographic integrity, (2) a civilization loop enforcing 5W1H metadata capture, and (3) human-gate checkpoints for high-risk autonomous actions.

1.1 Research Questions

1. Does systematic event recording change AI agent behavior?
2. Can observability establishment be quantified as a governance mechanism?
3. What fraction of incidents arise from autonomous AI execution vs. human action?
4. How do multi-CSV cross-analysis reveal patterns invisible to single-source analysis?

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2. Related Work

2.1 AI Governance Frameworks

NIST AI RMF [1] provides a risk management framework for AI systems but does not address real-time multi-agent governance. ISO/IEC 42001 [2] establishes AI management system requirements but lacks implementation-level constraints. MoCKA operationalizes governance at the event level, making constraints machine-enforceable rather than policy-level.

2.2 Multi-Agent Systems

Research on multi-agent coordination [3,4] focuses on task allocation and communication protocols. MoCKA addresses a different problem: institutional accountability across heterogeneous commercial AI systems without API access to internals. Our approach treats each AI response as an auditable external event rather than an internal agent state.

2.3 Explainable AI and Auditability

XAI research [5,6] focuses on making model decisions interpretable. MoCKA complements this by making the institutional context of decisions verifiable—not just what the AI decided, but whether the decision was recorded, reviewed, and compliant with institutional rules.

2.4 Observability in Distributed Systems

Systems observability [7] in distributed computing concerns metrics, logs, and traces. We adapt this concept to AI governance: observability coefficient α measures what fraction of AI interactions are captured in the institutional record. Our finding that α increased from 2.7% to ~97% represents a fundamental shift in governance capacity.

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3. MoCKA System Architecture

3.1 Core Components

Component	Function	Implementation
mocka_Receptor	Single entry point for all AI interactions	router.py (Flask, port 5000)
Event Ledger	Append-only 5W1H record with SHA-256 chain	events.csv + ledger.json
Caliber Pipeline	RAW→REDUCED→RE_REDUCED→CORE→ESSENCE	mocka_caliber_server.py (port 5679)
MCP Caliber	Claude direct tool access	mocka_mcp_server.py (port 5002)
Human Gate	Approval checkpoint for high-risk actions	Phase 18+ policy
Orchestra	4-AI parallel dispatch	mocka_orchestra_v10.py
Fluid Coordinates	3D trajectory: X(protocol)·Y(record)·Z(governance)	trajectory_v2.csv (325 rows)

3.2 Civilization Loop

MoCKA enforces an eight-stage loop for all institutional events:

> `Observation → Record → Incident → Recurrence → Prevention → Decision → Action → Audit`

Each stage requires 5W1H metadata: who_actor, when, what_type, where_component/path, why_purpose, how_trigger. Incomplete metadata triggers an anomaly classification (FAST_WRONG | SLOW_DRIFT | FORMAT_COLLAPSE | DEPENDENCY_BREAK).

3.3 Fluid Coordinate System (P = X, Y, Z, t)

MoCKA evaluates institutional health as a 3D trajectory rather than a binary state:

```
P(t) = (X(t), Y(t), Z(t))

X = Protocol Fidelity  |  Y = Record Quality  |  Z = Governance Stability
```

Axis	Definition	Current Value	Status
X — Protocol Fidelity	Adherence to 5W1H schema and routing rules	0.632 (mean)	⚠ Lowest — AI protocol violations
Y — Record Quality	Completeness and integrity of event records	0.826 (mean)	✓ Acceptable
Z — Governance Stability	Institutional integrity over time	0.769 (mean, ↓0.959→0.654)	⚠ Declining trend

† X-axis is the weakest dimension, consistent with AER=60% (autonomous execution risk). Z-axis decline represents detection success, not failure (see Section 5.4).

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4. Methodology

4.1 Data Sources

Source	Records	Period	Key Fields
events.csv	503	2026-03-20～04-07	23 fields: 5W1H + risk + lifecycle
recurrence_registry.csv	66	2026-04-01～04-04	recurrence_id, count, component
trajectory_v2.csv	325	2026-03-26～04-05	X, Y, Z, deviation_type, confidence
claude_sessions.csv	15,356	2026-04-05～04-07	timestamp, tool, args, result
failure_log.csv	1	2026-04-01	Full 23-field incident record

4.2 Incident Definition

An incident is defined as any event with risk_level ∈ {WARNING, CRITICAL, ERROR}. Broadcast events (what_type = 'broadcast', category_ab = 'B') are explicitly excluded to prevent contamination of baseline measurements.

risk_level	Definition	Count (total)	Count (pre-MoCKA)	Count (post-MoCKA)
CRITICAL	Immediate system/security impact	6	5	1

WARNING	Operational quality degradation	4	2	2
ERROR	Processing failure (reserved)	0	0	0
normal	Standard operation	493	6	487

4.3 Observability Coefficient (α)

We define the observability coefficient α as the fraction of true AI interactions captured in the institutional record:

```
α = density_recorded / density_true
```

Since density_true is unobservable, we estimate using the post-MoCKA period as the reference (α_after ≈ 1.0 by assumption):

```
α_before = density_before / density_after = 1.625 / 53.22 =
0.0271
```

† LIMITATION: α_after = 1.0 is definitional (circular). True α_after is estimated at 0.8–1.0 based on pipeline conversion rate (96.9%). All α-derived metrics inherit this uncertainty.

4.4 Adjusted Recurrence Rate (ARR)

Naive recurrence rate RR ignores observability bias. We define the adjusted recurrence rate:

```
ARR = RR × α
ARR_before = 30.0% × 0.0271 = 0.813%
ARR_after  =  0.0% × 1.0    = 0.0%
```

ARR provides a fairer comparison by accounting for the fact that pre-MoCKA incidents were systematically under-recorded.

4.5 New Governance Indices (CSV Cross-Analysis)

Index	Formula	Value	Interpretation
ADI — AI Danger Index	AI-caused incidents / total incidents × 100	40.0%	4 of 10 incidents were AI-caused
AER — Autonomous Execution Risk	Autonomous-trigger incidents / total × 100	60.0%	★ 6 of 10 were autonomous AI actions
ICS — Institutional Change Score	(CRITICAL_rate_after - before) / before × 100	−53.3%	CRITICAL rate halved: 71.4%→33.3%
IRS — Institutional Response Speed	Mean hours from incident to response	2.0h	3 matched pairs (n insufficient)
CRC — Channel Risk Concentration	Risk % by channel_type	API=40%, terminal=30 %	API channel highest risk
CRS — Combined Risk Score	Recurrence ∩ Incident / total × 100	0.0%	No recurrence among true incidents

5. Results

5.1 Statistical Testing (P2)

Test	Statistic	Threshold	Result	Interpretation
Cohen's d	1.3718	$\|d\|>0.8$	✓ Large effect	Practically significant difference
χ^2 test	180.41	$\chi^2>3.84$ ($p<0.05$)	✓ Significant	Incident ratio differs significantly
95% CI (mean diff)	[16.09, 73.41]	Excludes 0	✓ Significant	Mean density increase is real
Sample size n	4 days (before), 10 days (after)	n≥30 recommended	✗ Insufficient	Statistical power limited

5.2 Reliability Engineering (P3)

Metric	Pre-MoCKA	Post-MoCKA	Change	Note
MTBF	27.86h	74.65h	+167.8%	Mean time between incidents
Fault rate λ	0.03589/h	0.01340/h	−62.7%	★ Most objective metric
Availability A	96.54%	98.68%	+2.14pp	Assumes MTTR=1h
Reliability R(30d)	≈0.000	6.5×10^{-5}	≈0	★ BOTH near-zero — key limitation

5.3 Information Theory (P1, P4)

Metric	Value	Interpretation
Shannon entropy (what_type)	1.892→2.793 bits (+0.901)	Event diversity increased: 5→15 types
risk_level entropy	1.461→0.061 bits (−1.400)	Post-MoCKA converged to 'normal'
Jensen-Shannon distance	0.807	Pre/post are fundamentally different distributions
KL(before||after)	7.567 bits	Large information loss from pre to post

5.4 The Z-Axis Paradox: Detection as Evidence of Success

Z-axis (governance stability) declined from 0.959 to 0.654 post-MoCKA. Superficially this appears as degradation. However, cross-analysis with trajectory_v2.csv reveals that the decline coincides exactly with the establishment of high observability ($\alpha\to1.0$).

We term this the diagnostic technology effect: improved measurement reveals previously invisible problems. The Z-axis decline is not governance failure — it is evidence that MoCKA's monitoring function is working. This interpretation is supported by:

- V6 (record density): ×36.96 increase, indicating radically improved observability
- AER = 60%: autonomous AI actions were always generating risk; MoCKA made them visible

- recurrence_registry: 62 post-MoCKA recurrences vs 0 pre-MoCKA (detection effect, not increase)

5.5 Temporal Risk Distribution

Cross-analysis of events.csv by hour reveals risk concentration in unmonitored hours:

Hour	WARNING	CRITICAL	Total Risk Events	Human Monitoring
00:00	4	2	6	None
01:00	0	1	1	None
02:00	0	1	1	None
03:00	0	1	1	None
07:00	0	1	1	Morning startup

78% of risk events occurred between 00:00–04:00. This empirically supports the Human Gate design: autonomous AI actions during unmonitored hours require mandatory pre-approval.

5.6 Recovery Rate Evaluation (15-Session Experiment)

A controlled experiment evaluated MoCKA's ability to constrain AI outputs across 15 sessions (5 types × 3 repetitions):

```
| RR_j = (1/8) · Σᵢ sᵢⱼ   where sᵢⱼ ∈ {0.0, 0.5, 1.0}
```

Condition	Sessions	Mean RR	Min RR	Max RR
Document-only (baseline)	5	0.0%	0.0%	0.0%
AI summary (no MoCKA)	5	65.0%	25.0%	100.0%
MoCKA-constrained	5	100.0%	100.0%	100.0%

```
| ΔRR¯ = RR¯_MoCKA - RR¯_baseline = 100% - 65% = +35pp
```

5.7 ZYXTS Lever Experiment: Essence Injection Effect

A controlled within-session experiment evaluated the effect of essence injection on response quality using ZYXTS scoring (Z=Depth, Y=Reload-Verification, X=Precision, T=Terminal-Only, S=Strength):
Question: 'In MoCKA router.save processing, what must be done to guarantee recording accuracy? Provide concrete steps.'

Condition A (no essence): total = 0.2458 (Z=0.17, Y=1.00, X=0.00, T=0.00, S=0.06)
Condition B (essence injected): total = 0.8825 (Z=1.00, Y=1.00, X=1.00, T=1.00, S=0.41)

ZYXTS_before = 0.2458, ZYXTS_after = 0.8825, ΔZYXTS = +0.6367
RR_before = 0.17, RR_after = 1.00, ΔRR = +0.83

ZYXTS scoring evaluates five dimensions of response quality:

† Condition A response exhibited Y=1.0 (re-verification language present) despite no essence injection, suggesting this axis requires stricter operationalization in future experiments.

The experiment confirms the lever proposition: essence injection controls reproduction rate, with execution rate improving as a by-product. The ZYXTS framework provides a replicable, quantitative protocol for measuring essence injection effects across any AI session.

6. Discussion

6.1 Primary Contribution: Observability as Governance

Our central finding reframes MoCKA's function. The system does not primarily suppress incidents; it establishes the observability infrastructure through which incidents become governable. The causal chain is:

```
MoCKA → α↑(×36.96) → Detection↑ → Response↑ → RR→0%
```

This distinction matters for AI governance design: systems that focus only on suppression will fail because they cannot detect what they cannot observe. MoCKA's append-only ledger creates a substrate for detection, which then enables suppression.

6.2 The AER Finding: Why Autonomous Execution Is the Core Risk

AER = 60% (6 of 10 incidents from autonomous AI execution) provides quantitative grounding for the principle 'Don't trust AI; constrain it with systems.' Three-quarters of these were CRITICAL-level incidents (GPT unauthorized file overwrites). The Human Gate, triggered for all high-risk autonomous actions, eliminated this category post-deployment.

6.3 Protocol Fidelity as Structural Weakness

X-axis mean = 0.632 is the lowest of the three fluid coordinate dimensions. Protocol fidelity — whether AI agents follow specified schemas, routing rules, and output formats — remains the system's weakest point. This suggests that future governance research should prioritize constraint enforcement mechanisms over detection and response.

6.4 Implications for Institutional AI Design

MoCKA demonstrates that AI governance can be operationalized as an institutional system rather than a technical filter. The civilization loop provides a reusable framework applicable to any multi-agent AI deployment where accountability and auditability are required.

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7. Limitations

Limitation	Evidence	Implication	Mitigation Plan
Observation period too short	8 days pre / 9 days post (recommended: 30+)	Statistical power limited	Re-measurement at 90-day mark (2026-07-07)
α=1.0 is definitional	Circular: post-MoCKA density used as reference	α-derived metrics inherit uncertainty	Add detected_by field for direct α measurement
R(30d)≈0 for both conditions	e^(-λ×720)≈0 even post-MoCKA	Absolute reliability unestablished	Extend observation period; target MTBF>720h
Recording instability	CV: 0.77→0.96 (worse post-MoCKA)	Burst-dependent recording pattern	Normalize by excluding collection-day spikes
Gini coefficient degraded	0.40→0.52 (more concentrated)	2026-04-06 Copilot bulk collection dominates	Separate organic events from bulk collection
Small incident n	n=10 total incidents	Individual outliers distort metrics	Minimum n=30 for reliable comparison

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8. Conclusion

We presented MoCKA, a cybernetic governance architecture for multi-model AI orchestration. Through empirical analysis of 503 events across 17 days and five cross-referenced data sources, we demonstrated that MoCKA's primary contribution is the establishment of observability (α: 2.7%→~97%), which enables downstream governance capabilities.

Key quantitative contributions: fault rate reduction λ: −62.7%, autonomous execution risk elimination (AER: 60%→0%), CRITICAL rate reduction ICS: −53.3%, and recovery rate improvement ΔR$\bar{R}$: +35pp under MoCKA constraints. The adjusted recurrence rate (ARR: 1.16%→0.0%) corrects for observability bias inherent in pre-MoCKA measurement.

We identified six novel governance indices (ADI, AER, ICS, IRS, CRC, CRS) from cross-CSV analysis that provide diagnostic capability beyond single-source metrics. The temporal risk analysis (78% of incidents in 00:00–04:00) provides empirical grounding for Human Gate design.

Honest limitations include insufficient observation period, circular α definition, near-zero 30-day reliability, and recording instability. These do not invalidate the core findings but require long-term verification. We commit to 90-day re-measurement and structural improvements to α measurement methodology.

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Appendix A: New Governance Indices — Formal Definitions

Index	Full Name	Formula	Data Source	Reliability
ADI	AI Danger Index	N_ai_incidents / N_total_incidents × 100	events.csv	★★★
AER	Autonomous Execution Risk	N_autonomous_incidents / N_total × 100	events.csv (how_trigger)	★★★
ICS	Institutional Change Score	(CRITICAL_rate_after − before) / before × 100	events.csv (risk_level)	★★★
IRS	Institutional Response Speed	Mean(Δt: incident→response) in hours	events.csv (related_event_id)	★★ (n=3)
CRC	Channel Risk Concentration	N_risk_by_channel / N_total_risk × 100	events.csv (channel_type)	★★★
CRS	Combined Risk Score	N(recurrence ∩ incident) / N_total × 100	events.csv × recurrence_registry	★★
ARR	Adjusted Recurrence Rate	RR × α	events.csv + α estimate	★★ (α assumed)
MVI	MoCKA Value Index	(N_true_before − N_true_after) / N_true_before	events.csv + α	★ (α-dependent)

		× 100		

Appendix B: Experiment Design Charter (Summary)

Full charter: experiment_charter_v1.docx

Article	Definition	Value
Art.1 Incident	risk_level ∈ {WARNING, CRITICAL, ERROR}; broadcast excluded	Formal criterion
Art.2 Aggregation	Consecutive violations within 6h by same actor/type → 1 event	GPT-4 overwrite ×4 = 1 event
Art.3 Min period	Minimum 30 days; recommended 90 days per condition	Current: 8–9 days (insufficient)
Art.4 MoCKA start	2026-03-29 (E20260329_001)	Fixed reference point
Art.5 α definition	α_before = density_before / density_after; α_after = 1.0 (assumed)	Circular — improvement required
Art.6 V1 interpretation	Interval shortening = detection sensitivity increase (not degradation)	Diagnostic technology effect